

ECOLOGY AS COGNITIVE ARCHITECTURE

Learning as cost-accounted computation in a reflective higher-order calculus —
and what happens to the unit of computation when a single learner is not enough.

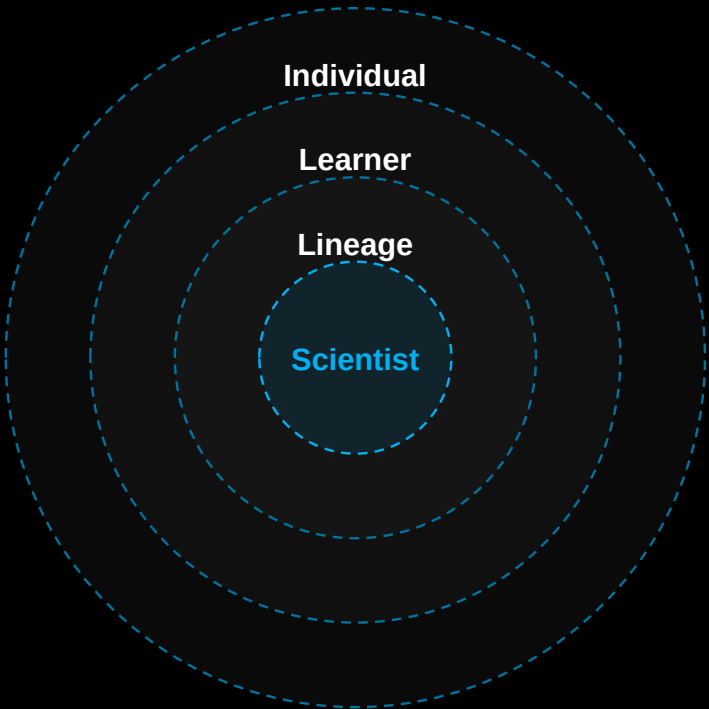
The mortal scientist · learning to play · composing learners · engines and individuals

Lucius Gregory Meredith | CEO, F1R3FLY.IO · CHIEF SCIENCE OFFICER, F1R3FLY INDUSTRIES



THE UNIT OF COMPUTATION KEEPS MOVING

Four notes, four units. At each level a result says the unit cannot get where it needs to go, and the operator that escapes is supplied only by the level above.



EACH LEVEL IS CONFINED BY A RESULT PROVED AT THAT LEVEL		
Mortal scientist	confined below 2^{-n}	escapes by crossover
Lineage	one root only	escapes by parallel composition
Learner	no scale structure	escapes by engines, then merger
Individual	bounded by r^*	escapes by becoming a scientist

The rings are dashed on purpose. Perfect closure is death: a computation that cannot be reached cannot harvest and cannot assay.

Same sort, same rewrite rule, one conservation law. The unit expands; the calculus does not.

WHAT DOES IT TAKE FOR A COMPUTATION TO DO SCIENCE?

Not fitting a function to a supplied dataset. There is no dataset — the learner must go and get its observations, getting them costs, and getting them changes the source.

R1

An environment

...and a ceiling on what can be known of it. Without a ceiling there is no saying whether a learner is doing well.

R2

A hypothesis language

Fine enough to express every distinction the learner could observe, and no finer.

R3

Hypothesis → verdict

What an experiment is, how a hypothesis determines one, and what comes back — including nothing.

R4

A motive

If an external scorekeeper hands out rewards, the model has outsourced the thing it set out to explain.

Three of the four answers turn out not to be choices. The first note is largely the discovery that we had none left.

COMPUTATIONS ARE ONTOLOGICALLY ISOLATED

So the environment of a computation is other computations, and the ceiling on what one can learn of another is derived rather than posited.

THE CUT

W $\equiv$ **S** | **E**

What S can learn of E is bounded above by E's context-labelled bisimulation class.
That is the ceiling R1 asks for.

THREE GRADES OF ACCESS — AND ONLY THREE

Perception

structural predicates at the surface — priced by depth

Interaction

$\langle K_i \rangle \phi$ — build a probe, plug, observe: a rendezvous, and time

Reflection

@ ϕ on a name you already hold — cheap, no communication

OBSERVATION VERSUS DISCLOSURE

Behaviour is what one observes across a cut; structure is what one is given, or takes. Perception reads the surface of things one does not hold; reflection reads the interior of things one does. The operation that converts the second into the first is eating — and it is fatal to the thing converted.

THE LOGIC IS GENERATED, NOT DESIGNED

Feed the rho calculus' own presentation (Σ , E , R) to OSLF and it emits namespace logic. Formulae denote bisimulation equivalence classes.

TOO COARSE

Distinctions the learner can observe but cannot state. Blind in a way its own faculties do not require.

ADEQUATE

The formulae separate exactly what interaction separates. Neither failure is possible.

TOO FINE

Hypotheses that differ on environments no experiment could separate. Tokens — its life — spent on nothing.

ADMITTING THE STRUCTURAL PREDICATES IS WHAT GIVES THE LEARNER SENSES

A sense is a structural predicate evaluated at the interaction surface — one sees the skin of a rock, not its interior. Seeing deeper costs more, so the interior is not free, merely expensive, and past a point unaffordable.

$|_k$ **out(n)φ** **in(n)φ** **⟨K_j⟩φ** **@φ**
νX.φ

The hidden-name quantifier H is gone: freshness by quotation makes @φ enough.

Acuity is not an endowment — it is whatever the scientist can currently pay for.

A designed logic has no reason to be adequate. A generated one is adequate whenever its target specification carries the connectives adequacy needs.

A HYPOTHESIS IN GUARD POSITION IS AN EXPERIMENT

The for-comprehension carries a where clause, so a formula sits in guard position. The checker is the reduction rule; a confirmation is a rendezvous.

THE ASSAY — A COMPLEMENTARY GUARD PAIR

```
Kj[ ... ]
| for (x <- p where phi(x)) { confirm!(*x) }
| for (x <- p where ~phi(x)) { refute!(*x) }
```

The negation is what makes this more than an expression of hope: a guard that fails to fire is otherwise indistinguishable from one still waiting.

THREE OUTCOMES, NOT TWO

CONFIRM	the prediction held
REFUTE	the prediction failed
$\perp$	nothing came back before the budget ran out

$\perp$ is not a defect. It is the honest residue — and separating it from falsity is the whole gain. A mortal scientist can only run assays decided within its budget, so the fragment of the ideal logic it can test is bounded in modal depth. That fragment is not adequate.

A finite scientist necessarily holds a non-adequate theory. Which fragment it holds is fixed by what it can pay for — the budget is the inductive bias.

NOBODY ADJUDICATES. THE BAD PREDICTOR STARVES.

Tokens are conserved and never minted. A hypothesis says where the tokens are, and the reward for a good one is the tokens.

CONSERVATION

$$\sigma + \kappa = \theta$$

Spent tokens do not vanish; they migrate into the record. The accumulation of κ is the arrow of time, and science is entropy production.

EATING IS A RECEIVE

for (t <- m)

A stack is a message at a metabolic channel.
Harvest is an ordinary receipt, death is failure to fund the next step, defence is restriction.

PREDATORS ARE NON-IMAGE CONTEXTS

c $\notin$ $\llbracket \mathcal{C}(\rho) \rrbracket$

In the decorated calculus a stack is inviolable. In its image it is a message anyone holding the name may take. The ecology lives in that gap.

Vacuity is punished by hunger rather than by rule: T predicts nothing, locates no metabolic channel, yields nothing. And Occam's razor is metabolic — description length and food are denominated in one currency, so MDL is derived rather than stipulated.

THE HYPOTHESIS SPACE IS AN ULTRAMETRIC TREE

Balls are nested or disjoint. Every point of a ball is its centre. There is no betweenness — so there is no gradient, and gradient methods are not a poor fit but ill-typed.

CONFINEMENT

$$d(\varphi_0, \varphi_k) \leq \max_i d(\varphi_{i-1}, \varphi_i)$$

A revision walk whose steps are all shorter than 2^{-n} never leaves the ball of radius 2^{-n} it began in. Not slowly — at all. Whatever ball a scientist's early commitments place it in is the ball it dies in, unless it makes a move that is not small.

TWO DIALS, AND THE PRICE OF EACH

Distance

= backtracking depth. A move of 2^{-n} discards every confirmation bought at depth $\geq n$.

Direction

= a choice of sibling. Which structural predicate to vary; the branching is fixed by the connectives OSLF emitted.

CHEAP MOVES ARE CONFINED · UNCONFINED MOVES ARE DEAR

Strategies: conservative (cheap, confined) · radical (unconfined, expensive) · adaptive, with the refutation rate as temperature — a control parameter the assays already emit as messages. A mortal scientist is not rewarded for true hypotheses but for nutritive ones, and the branching it can survey is bounded by its wealth. Poverty is confining.

THINGS WE DID NOT PUT IN

Each follows from the commitments rather than being added to reach it.

1

DESTRUCTIVE ASSAY FORCES A LOGIC OF NAMESPACES

Harvest ends the prey's computational life, so each specimen affords exactly one measurement and a hypothesis about an individual is worthless after its own first test. Only class-level hypotheses pay. Induction is not an epistemic virtue the scientist displays — it is the only strategy that survives its own testing.

2

METABOLIC VIABILITY LOWER-BOUNDS OBSERVABILITY

The perfect defence is to close: a fully restricted computation cannot be perceived, so its metabolic channel cannot be derived. It also cannot eat. Any computation that survives indefinitely must expose surface, hence must be perceivable, hence is in principle edible. Perfect concealment is available only to the starving.

3

INFORMATION OPPOSES SURVIVAL

Perceive: nothing learned about the interior, nothing destroyed. Assay: perturbs. Break in: fatal, and the only rung that pays. The better the theory, the faster it exhausts the environment that confirms it — success is self-terminating.

The apparatus of the scientific method stops looking like something a mind does, and starts looking like something a mind, so situated, could hardly avoid.

A SOURCE AND A PREY DIFFER BY ACCESS PROFILE

The sun is a finite reservoir in a system where energy is conserved. Openness is a bookkeeping artefact of where the boundary was drawn — so conservation is the picture at the correct scope.

	A SOURCE	A PREY
<i>release</i>	rate-limited; cannot be forced	a single linear send
<i>exhaustible</i>	no	yes
<i>exclusive</i>	no — many may harvest	yes — a linear receipt
<i>adaptive</i>	no	yes, and it models you back
<i>persistent emitter vs. linear send — same sort, same rewrite rule, different multiplicity</i>		

THE TWO SCIENCES

A source is a structurally regular, non-adaptive recursion, so its complete theory is exact at a finite unfolding: the source-feeder's inquiry converges and its epistemic outlay falls to rent on a fixed formula.

A prey is a mortal computation that revises against you. No finite theory is ever complete, and the cost is permanent.

INTELLIGENCE IS A HETEROTROPH'S EXPENSE

Plants do not need brains because the sun is the environment whose complete theory is affordable. That this is also true of the Earth we take as corroboration of the modelling, not as its source.

THE GERM IS A QUOTATION

@Code does not reduce: it does not metabolise, does not age, and burns nothing while held. *@Code runs, and the meter starts. Reflection is the germ line.

A LOCUS IS A CONTEXT

The normal form recurses, so a genome is a tree of parallel compositions whose edges are prefixes. A locus is the one-hole context K from root to hole — the same contexts the modality $\langle K_j \rangle$ already labels. Homology is agreement of channel path and binding environment, so reproductive isolation is namespace disjointness and a species is a formula.

THE UNIT OF HEREDITY IS A REDEX

Swap communicating pairs, not parands: a gene is an interaction, not a piece of structure, and linkage is derived rather than posited. Balance is preserved exactly when the swap is channel-degree preserving — so preserving races is the side condition that makes the guarantee hold, not a second benefit.

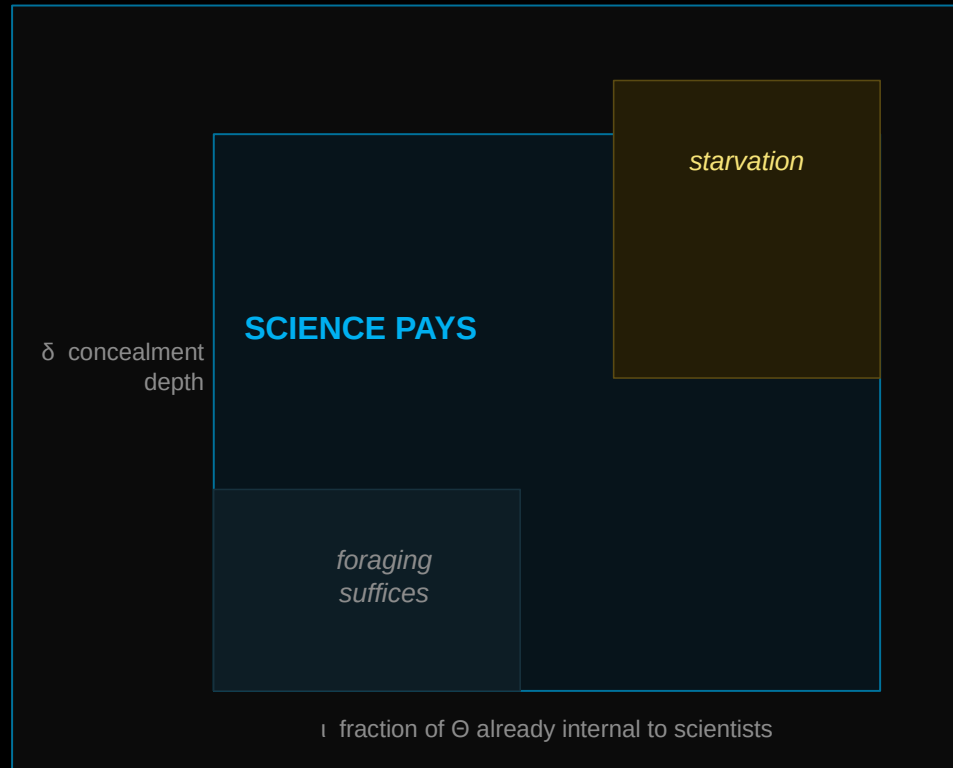
SELECTION IS POST HOC

Viability is a formula — closure, reading frame, balance, liveness — so screening an embryo is model checking, and model checking is priced. A parent cannot emit only viable offspring, not for want of a criterion but because applying it is unaffordable. Were checking free, parents would design and selection would have nothing to do.

Confinement is why this matters: a crossover at a shallow locus reaches distance 2^{-1} , which Proposition 13 denies any individual. The lineage is the second unit of learning.

SCIENCE IS A PHASE, NOT A UNIVERSAL ADVANTAGE

Θ is fixed, so a world is a partition of a conserved quantity and a distribution over configurations is a microcanonical ensemble — exactly, because conservation is exact.



THE AUDIT — WHAT THE ASSUMPTIONS ALREADY FIX

FORCED the hypothesis language · three grades of access and no fourth · assays as guarded receipts · budget-relative refutation · hypotheses about namespaces · eating as an ordinary receipt · death as failure to fund · $\sigma + \kappa = \sigma_0$ · predators as non-image contexts · exposure · information against survival · a locus is a context · name growth bounded by term size · selection post hoc

CHOSEN Θ · the initial distribution · the admitted context class · the pricing schedule · concealment depth · provisioning · the crossover choice function · the revision policy

The physics is forced. The ecology and the psychology are the free parameters.

WHY A SOLVED GAME

Noughts-and-crosses, built as an energy landscape: an arena that holds no stack and merely routes, a stationary practice partner that is a source, and a population of scientists whose hypotheses are formulae in the generated logic.

1

A MOVE IS AN ASSAY

Playing and experimenting are one act. The guard pair sits on the move channel, so a ply is a rendezvous that either confirms, refutes, or times out against the budget.

2

MINIMAX IS AN ALTERNATING MODAL FORMULA

So the companion note's asymmetry between existential and universal modalities predicts that attack is learned before defence — and the measurement finds it.

3

A FORK IS GRADED SEPARATION AT $K = 0$

Threat $\mid_0$ Threat. The concept game programs normally hand-code is a term the generated logic already contains. Requiring it forces the squares, not the lines, to be the parands: two threats with a common gap must be excluded.

Three things fall out of the mapping rather than being put into it. The third dictates the board encoding.

THE REDEX CONTEXT IS THE MOVE

K_j must be a redex context, so an empty square is a receipt offering to be claimed and an occupied one is a resting send. A ply is exactly one rewrite, and $\langle K_s \rangle$ is a generated primitive rather than a derived abbreviation.

MOVES ARE REDEX POSITIONS

Redex positions correspond bijectively to empty squares, so the branching factor of $\exists s$ is the legal move count — which is exactly what the bounded-branching proposition needs.

WHY AN ACTION LABEL WOULD NOT DO

Every claim is the same action. An action-labelled HML sees one label and cannot distinguish nine moves; the context-labelled modality sees as many labels as there are enabled interactions.

PERCEPTION MUST NOT BE ABLE TO ACT

Threat offers live on their own names. A line signalling on a move channel would rendezvous with the square's own receipt and play the move. Grades 1 and 2 are separated by a choice of namespace.

TWO ROUTES TO ANY PREDICATE

Read it — perception, priced by formula depth — if the environment has already computed it into legible structure; or drive it — interaction, priced by rendezvous. Putting the threats into the term does not remove the cost, it relocates it from the scientist's budget to the world's. The interesting question is who is billed, and the framework can pose it because both parties have stacks.

THE LADDER PAYS, AND THE TRUE THEORY DOES NOT

Seven rungs priced by the modal depth of the formula each installs, computed exactly over all 255,168 plays. Rungs 1, 2 and 5 are non-modal; 3 and 4 are the modal ones.

RUN G	WHAT IT INSTALLS	WIN	COST	MARGINAL RETURN
0	random play	.437	0	—
1	win now (perception)	.668	40.9	+.0057
2	block now (perception)	.798	77.8	+.0035
3	$\exists s. \langle \langle K_s \rangle \rangle \text{Fork}$	.831	134.9	+.0006
4	$\exists s \forall t. \langle \langle K_s \rangle \rangle [K_t] \neg \text{Fork}$	.836	245.7	+.00005
5	Central — spatial, depth 0	.918	255.9	+.0082
χ	$\forall X. \forall s. [K_s](\neg \text{Lost} \wedge \exists t. \langle \langle K_t \rangle \rangle X)$	.873	679.2	−.0001

WHERE THE YIELD ACTUALLY IS

92.3%

of the total gain is delivered by the three non-modal rungs — for 34.3% of the cost.

The spatial rung — the one a purely behavioural logic could not state — returns +.0082 per metered unit, the best purchase anywhere. Rung 4, the single genuinely alternating rung, is the worst: +0.005 for 111 units.

χ is dominated: it wins less often than the heuristic rung beneath it while costing well over twice as much. Truth buys safety, not food.

DEEPENING THE THEORY MAKES THE OPENING WORSE

The best opening reverses with theory depth. A lineage whose theory is deep but whose opening is fixed cannot repair the opening by deepening.

AT RUNG 5

centre .833 corner .597

With a heuristic theory the centre opening is the better one, as every beginner is taught.

AT X

corner .896 centre .548

At the exact characteristic formula the ordering reverses. The received opening becomes the worse one.

HOLDING THE OPENING FIXED

Deepening from rung 5 to χ while keeping the centre opening takes the win rate from .833 down to .548. The two loci are reachable to a lineage and not to a learner walking the lattice.

AND A SOLVED SCIENCE IS A DEAD ECOLOGY

χ -monoculture and centre-rung-5 monoculture both draw every game, so the harvest rate is 0.000 and the population starves on a finite wall. The corner-rung-5 population sustains .333 and lives. Imperfection is load-bearing: the better the science, the sooner the world it feeds on is exhausted.

The corner-opening mutant invades at identical cost, never losing as first player. It is exactly the distance- 2^{-1} move that confinement denies an individual — and a lineage gets it free.

A POPULATION THAT SOLVES ITS GAME STOPS EATING

Conservation forces it. Nothing is minted, harvest requires a prediction that locates a stack, and a world in which every prediction is right is a world in which nothing is left to find.

THE HEAT DEATH

Every member holds a theory that draws against every other. Harvest rate falls to zero, σ declines monotonically, and the population burns tokens simply to remain alive.

WHY IT IS NOT AN ARTEFACT

It does not depend on the game being small or the pricing schedule being right. It depends only on conservation and on the fact that harvest is a receipt that a correct prediction locates.

WHAT IT COSTS THE FRAMEWORK

Nothing — but it does say the single-game world is the wrong scope. The heat death was never a property of noughts-and-crosses. It was a property of a world with one game in it.

Three exits are available: enlarge the game, let the rules drift, or admit an external source. The next note takes the first — and finds it was an operation inside the framework all along.

A LEARNER IS A NAMESPACE

Take the population as the learner: a distribution over scientists, over mortal computations holding no hypothesis, and over resource reservoirs. Because that population is carried by names manufactured by quotation, it is described by a name predicate rather than enumerated.

1

POPULATION

Scientists, non-scientist mortal computations, and stacks. The mixture is what the first note already assumed; here it is named as an object.

2

NAMESPACE

Rooted at the learner's own quotation, so membership is a formula. Reproductive isolation, species, and modularity all become statements in the same language.

3

DISTRIBUTION

Over configurations of the above, and inside the learner. Learners then compose by parallel composition, and the whole content of the composition is what their namespaces share.

The distribution is part of the learner, not a decoration on it — which is what puts composition in the total space of the fibration rather than its base.

DISJOINTNESS IS EXACTLY FACTORISATION

One operator, and a theorem that turns a modelling caveat into a measurement.

THEOREM

namespace disjointness $\Leftrightarrow$ **no spanning redex** $\Leftrightarrow \mu_{1\ 2} = \mu_1 \otimes \mu_2$

So the graded separation $|_0^N$ is probabilistic independence, and composition is strictly monoidal at zero grade.

THE DEFECT IS THE INTERACTION

The lax comparison map $\lambda : s(L_1) \otimes s(L_2) \rightarrow s(L_1 \mid L_2)$ is an isomorphism exactly at $k = 0$. Everywhere else its failure to be iso is not a caveat about the construction — it is the interaction, as a computable quantity.

NON-INTERFERENCE IS NOT STABLE

Names manufactured independently may coincide, so disjointness can be lost without anyone intending it. Root each learner's namespace at its own quotation and freshness-by-quotation makes disjointness a theorem. Modularity is purchasable; the price is a naming discipline.

ONE GRADE IS NOT ENOUGH

A learner's namespace is typed — metabolic, source, probe, mating — and grading separately over each recovers the classical table of ecological relations as sign conditions rather than importing it.

$k = (k_{\text{met}}, k_{\text{src}}, k_{\text{prb}}, k_{\text{mat}})$ — later, a fifth: k_{chn}

one grade per typed namespace: what the two learners share of metabolism, of sources, of probes, of mating

NEUTRALISM

$k = 0$ everywhere. Independence, and the only case in which the composite factorises.

COMPETITION

Overlap in source or metabolic namespaces, with both signs negative. Delivered without contact.

PREDATION & THEFT

Metabolic overlap with opposed signs, or probe overlap taken without payment.

MUTUALISM

Positive on both sides — and available only over the probe namespace.

Same lesson as the board encoding, one level up: choosing N is the content. Symbiosis is metabolic pooling — the composite the trophic section already predicted.

COOPERATION IS EPISTEMIC; COMPETITION IS METABOLIC

Tokens are conserved. Formulae are not. That single asymmetry decides which ecological relations can be positive-sum and which cannot.

NO METABOLIC MUTUALISM

If the overlap is confined to the metabolic and source namespaces then $\Delta_1 + \Delta_2 \leq 0$. Whatever one learner gains in tokens the other has lost, because there is no minting anywhere in the system.

POSITIVE SUM LIVES IN THE PROBE NAMESPACE

A formula can be read by both parties without either losing it, and a written surface costs nothing to leave standing. Amortised perception is the mechanism: cost relocation, and the question of who is billed.

THE QUALIFICATION THE ENGINE NOTE ADDS

Once communities have distinct token colours, conversion between dead stock and live stock is positive-sum in residual life even though it conserves the underlying quantity. So the slogan survives in tokens and needs care in lifetimes — the accounting is exact, the welfare claim is not.

WHETHER TRUTH IS AFFORDABLE IS A PROPERTY OF THE ENVIRONMENT

Nim, laddered by the same method and the same pricing schedule as the game note — and it inverts that note's headline.

NIM(3,4,5) — RETURNS FLAT

win .5000 / .6567 / .8899 / .9966
cost 0 / 38.9 / 80.4 / 99.6
return +.00403 / +.00562 / +.00555

χ is not dominated here.

NIM(1,2,3) — RETURNS RISING

win .5000 / .5917 / .7958 / .9000
cost 0 / 15.2 / 31.4 / 38.5
return +.00602 / +.01262 / +.01476

χ is the best buy on the ladder.

COMPETITIVE DILUTION, IN CLOSED FORM

At a shared rate-limited source with shares $\propto 1/\text{cost}$, each learner retains exactly $c_{\text{other}} / (c_{\text{self}} + c_{\text{other}})$ of the inquiry it could afford alone — independent of every other parameter. The dilution is regressive: it costs the unfinished science 66.9% of its affordable depth and the finished one nothing.

Competitive dilution then delivers the loss without either learner perceiving, modelling, or touching the other. It is competitive exclusion in epistemic currency.

ENLARGING THE GAME AND COMPOSING ARE ONE OPERATION

A learner in the terminal state — harvest zero, χ held by every member, σ declining — recovers a satisfiable foraging inequality when composed with a learner whose probe namespace is generated from a distinct root.

WHY THE STACKS ARE VISIBLE AGAIN

The second learner's stacks are concealed at a depth the first has not paid to see. Paying is once more a purchase with a return, so the metabolic channel can be derived and harvest resumes.

COMPOSITION REACHES WHAT NO LINEAGE REACHES

A lineage can only install alleles that already occur in one of its members, so every namespace it can come to describe is generated from its own root. A change of root is unreachable by recombination and reachable by composition.

AND THE SERIES DOES NOT STOP HERE

Composition of learners yields a learner, so the construction is closed under an associative operator. But closure permits arbitrary nesting without selecting any levels — which is the gap the fourth note fills.

A science that has closed reopens by meeting a science that has not. No new rewrite rule, no new sort, and no intervention from outside the model.

THE ENGINE IS AN ORGANISM, NOT A CONTRACT

Give each community its own token colour and adjoin conversion contracts between colours. What arrives is not a new kind of thing but a fourth row in the access-profile table.

A FOURTH ACCESS PROFILE

Persistent and non-exclusive like a source.
Exhaustible per colour like a prey. And alone among the four in being adaptive in its rate. The analogue is not a decomposer but a mycorrhizal network.

THE COMBINATORS ARE ITS REACTION LANGUAGE

The financial combinators sit one level down, in cost-accounted form: a persistent issuer and a linear contract. One extension is needed — the observable that sets the rate reads the issuer's own inventory.

ARBITRAGE BECOMES A PROPERTY OF A MARKING

Because engines hold inventory, per-colour totals are conserved and the rate is the inventory ratio. The log-rate cochain is a function of the marking, so the Hodge split must be taken pointwise.

Its dissipation θ , posited as a parameter in the virtual-token work, is here endogenous: it is the metering charge of the rendezvous that implements the conversion.

DISTANCE IS A TERM, NOT A PARAMETER

Rho assumes perfect communication, and does so without loss of generality: a pathway is an interposed process. $\text{for}(y \leftarrow x_1)P \mid C(x_1, x_2) \mid x_2!(Q)$, where C may drop or corrupt.

A LADDER OF MEDIA

relay · lossy · corrupting · reading · converting. The engine is the top rung, so the fourth access profile is a rung rather than a separate kind of thing.

LOSS IS STRANDING, NOT DESTRUCTION

A dropped token is stranded rather than destroyed, so v is conserved and exergy is monotone. $\theta = \theta_{\text{toll}} + \theta_{\text{loss}}$ with $\theta_{\text{loss}} = -\log p$ — the second law acquires a mechanism.

THE TOWER

Media are processes, so they are nameable, so they form a namespace one level up the reflection tower. The medium namespace of one level is the endpoint namespace of the next. This is the generating step: the outside of one level is the inside of the next.

INDIVIDUALS ARE POROUS

An individual owns its internal media and does not own its boundary media, since retransmission requires the power to name what failed. Total closure would be fatal — no harvest and no assay — which is the individuation-level analogue of a solved science being a dead ecology. Integration depth is bought with reliability, and reliability is available only on a medium one can repair.

PRICES ARE SELF-CORRECTING EXACTLY WHERE THE ECOLOGY IS DENSE

The engine graph carries an error-correcting code on prices, and we can say precisely how strong it is.

$$\| \text{syndrome} \|^2 = a^2 (1 - R_{\text{eff}}(e))$$

detection strength of a single corrupted rate

$$\sum_e (1 - R_{\text{eff}}(e)) = \beta_1$$

Foster's theorem: the total detection budget is the first Betti number

THE BUDGET IS FIXED; THE GEOMETRY DISTRIBUTES IT

Total detection strength across all engines is exactly β_1 — the number of independent cycles. Adding engines does not create discipline; it reallocates a fixed amount of it.

SERIES ENGINES ARE CONFUSABLE

Two engines in series cannot be told apart by the code. Parallel ones can. Redundancy of pathway is what buys distinguishability, which is the graph-theoretic form of the same result.

AND MERGER DELETES THE CODE

Closing a region sets $\beta_1 = 0$ and destroys factorisation with it. Containment and pricing discipline are traded against each other, which is the tension the next slide resolves.

A bridge has effective resistance 1 and carries no pricing discipline at all — and bridges fail twice: no cycle to check them, and no owner to repair them.

ERROR CORRECTION IS WHAT INDIVIDUATES

With name collision as the error channel and re-rooting as repair, a region may close over its namespace — merge — exactly when its isoperimetric ratio clears a threshold.

$h(B) \geq (\varepsilon \theta / \rho) \cdot r$
linear, population exponential

polynomial growth $\Rightarrow r^* = \sqrt{(\rho / \varepsilon \theta)}$

an expander $\Rightarrow r^*$

THE STRUCTURAL FORM: PARTIAL CLOSURE

An individual is one whose internal medium namespace is closed and whose boundary medium namespace is open. Closure is what makes repair possible; porosity requires redex pathways, and a boundary channel without one is a stranding site.

DEEP INTEGRATION IS EXPONENTIALLY FRAGILE

Reliability across d hops goes as p^d , so integration depth is purchased. At $p = .999 / .99 / .95 / .90 / .75$ the optimal depths are $5 / 5 / 3 / 3 / 2$ and the net values $86.4 / 82.1 / 69.0 / 57.8 / 38.2$.

LEVELS ARE FORCED, NOT PERMITTED

A population exceeding $|B(r^*)|$ cannot be maintained as one closed namespace, so the only way to keep scaling is replication of r^* -sized units composed with engines. The hierarchy is a sequence of critical radii.

Ownership is what buys depth, so the definition of an individual is not a stipulation. It is a fixed point, and the circularity is the fractality.

LEVELS ARE A SHUTOFF SEQUENCE

Price the maintenance of each type of overlap. As θ rises the components of the grading vector extinguish in a fixed order, and each extinction is a level boundary.

BOUNDARY	OPERATOR	SURVIVING OVERLAP	RELATION
within a skill set	merger	all	one budget, one namespace
organism / organism	+ engine	Met (transfer), Src, Prb, Mat	kin, trade, culture
deme / deme	+ engine	Src, Prb, Mat	competition, culture
species / species	+ engine	Src, Prb	predation, eavesdropping
biosphere / biosphere	only	Prb	signals
beyond d*	—	none	nothing

AN ORCA

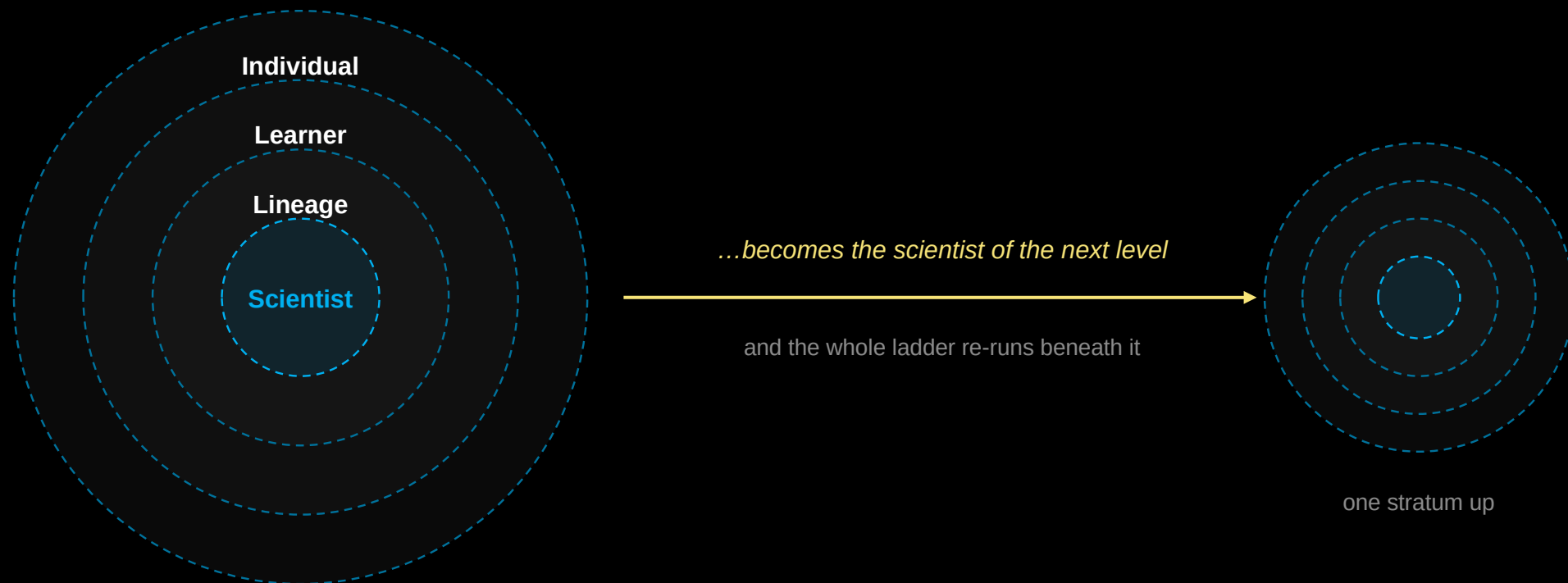
Hunting, place in the pod, mating, care of young: each skill backed by an ecology-as-learner reasoning through its own subdomain.

The orca's boundary media — calls, touch, the water — are the pod's internal media, and the pod is an individual exactly to the extent that it can name and repair them. The tower of media and the tower of individuals are the same tower.

The sequence terminates where $\theta \rightarrow \infty$ and even the cheapest channel prices out. Probe overlap goes last — which agrees, independently, with the incommensurability theorem.

AN INDIVIDUAL IS A SCIENTIST

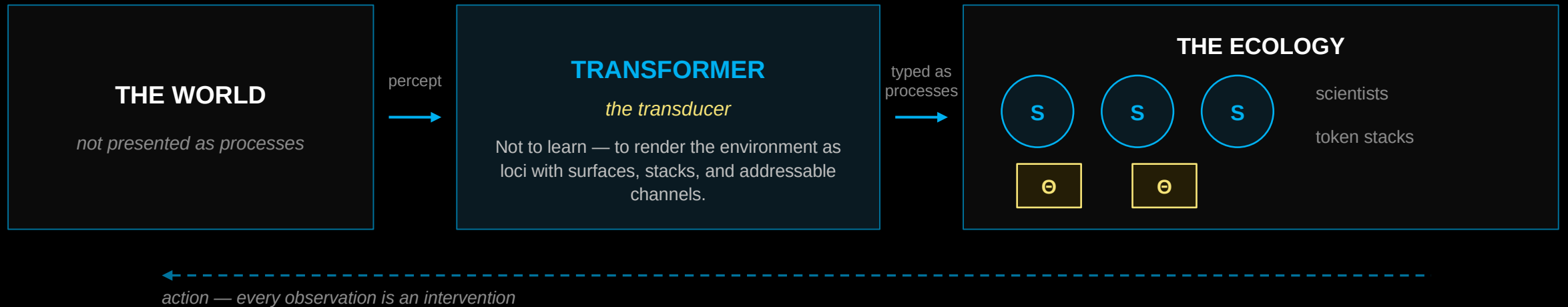
Not a fifth kind of thing. Being a scientist is a formula — a depth- ≥ 2 dialogue loop through a both-type channel — so the return edge is the one claim in this diagram a model checker could be handed.



Every boundary is a medium, owned by the ring outside it. The rings are dashed because perfect closure is death — and an individual satisfies the formula only if it stayed open.

TRANSFORMERS AS TRANSDUCERS

Posit a perceptual front end encoding the world as a population of computations and stacks. We are not proposing an interface — we are asking what follows if one exists.



THE NETWORK IS THE TRANSDUCER. THE ECOLOGY IS THE EPISTEMOLOGY.

Revision is local to a locus, so new learning has no mechanism by which to overwrite old. Exploration and exploitation are a budget, not a hyperparameter. And a population is required rather than preferred — a single learner is provably confined.

Wigner's question assumes a gap: mathematics a free creation, the world a brute given. There is no such separation here — the hypothesis language is generated by the same relation that fixes what can be observed at all. Mathematics is effective on exactly the domain interaction reaches, because it is the shadow the access relation casts.